



**Birla Institute of Technology & Science, Pilani**  
**Work Integrated Learning Programmes Division**

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**Digital Learning Handout**

**Part A: Content Design**

|               |                           |
|---------------|---------------------------|
| Course Title  | Deep Neural Networks      |
| Course No(s)  | AMLSIZG511                |
| Credit Units  | 4                         |
| Credit Model  | 3-1-0                     |
| Course Author | Prof. Seetha Parameswaran |
| Version No:   | 1                         |
| Date:         |                           |

**Course Description:**

Introduction to neural networks, approximation properties, back propagation, deep network training, regularization and optimization, convolution neural networks, recurrent neural networks, attention models, transformers, neural architecture search, federated learning, meta learning, applications in time series modelling and forecasting, online (incremental) learning.

**Course Objectives**

| No  | Course Objective                                                                                                                                                                                                                                                               |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CO1 | Apply deep learning techniques to solve complex problems in various domains, demonstrating proficiency in designing and implementing neural network architectures including feedforward, convolutional, and recurrent networks.                                                |
| CO2 | Analyze the performance and behavior of deep learning models, including the impact of depth, optimization challenges, and regularization techniques, to enhance model efficiency and effectiveness.                                                                            |
| CO3 | Evaluate the strengths and limitations of different deep learning architectures and techniques for specific problem domains, considering factors such as computational complexity, data requirements, and model interpretability.                                              |
| CO4 | Design and implement innovative deep learning solutions that integrate multiple techniques, such as attention mechanisms and neural architecture search, to address challenging problems in areas like computer vision, natural language processing, and time series analysis. |
| CO5 | Critically assess the latest advancements in deep learning, including transformers, federated learning, and meta-learning.                                                                                                                                                     |

**Text Book(s):**



|           |                                                                                                               |
|-----------|---------------------------------------------------------------------------------------------------------------|
| <b>T1</b> | Zhang, A., Lipton, Z. C., Li, M., & Smola, A. J. (2023). Dive into deep learning. Cambridge University Press. |
|-----------|---------------------------------------------------------------------------------------------------------------|

**Reference Book(s) & other resources:**

|           |                                                                                                                                                   |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>R1</b> | Goodfellow, I., Bengio, Y., Courville, A., & Bengio, Y. (2016). Deep learning (Vol. 1, No. 2). Cambridge: MIT press.                              |
| <b>R2</b> | Eugene, C. (2019). Introduction to deep learning. MIT Press                                                                                       |
| <b>R3</b> | Chollet, F., & Chollet, F. (2021). Deep learning with Python. Simon and Schuster.                                                                 |
| <b>R4</b> | Brownlee, J. (2018). Deep learning for time series forecasting: predict the future with MLPs, CNNs and LSTMs in Python. Machine Learning Mastery. |
| <b>R5</b> | Elsken, T., Metzen, J. H., & Hutter, F. (2019). Neural architecture search: A survey. Journal of Machine Learning Research, 20(55), 1-21.         |

**Learning Outcomes: Students will be able to**

|     |                                                                                                                                                                                                                                                         |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LO1 | Implement and optimize various deep neural network architectures (including feedforward, convolutional, and recurrent networks) to solve complex problems in areas such as image recognition, natural language processing, and time series forecasting. |
| LO2 | Analyze the impact of different hyperparameters, optimization algorithms, and regularization techniques on the performance of deep learning models, and apply this knowledge to improve model training and generalization.                              |
| LO3 | Evaluate the effectiveness of different deep learning architectures and techniques for specific problem domains, considering factors such as computational efficiency, data requirements, and model interpretability.                                   |
| LO4 | Design and implement advanced deep learning solutions that incorporate attention mechanisms, transformers, or neural architecture search techniques to address challenging real-world problems.                                                         |

**Prerequisites**

AIML\* ZC416 Mathematical Foundations for Machine Learning

AIML\* ZC418 Introduction to Statistical Methods

AIML\* ZG565 Machine Learning

**Modular Content Structure**

1. Fundamentals of Neural Network
  - Supervised, unsupervised, semi-supervised and reinforcement learning problems.
  - Why Deep Learning?
  - Applications of Deep Learning
  - Perceptron and Perceptron learning algorithm
  - Multilayer Perceptron (MLP)
  - MLP as classifiers and Universal approximators
  - Issue of Depth and Width





2. Deep Feedforward Neural Networks
  - Forward and backward propagation
  - Computation graph
  - Gradient Descent algorithm
  - Impact of depth in DNN
3. Optimization of Deep models
  - Challenges in Neural Network Optimization – saddle points and plateau
  - Non-convex optimization intuition
  - Overview of optimization algorithms
  - Momentum based algorithm
  - Algorithms with Adaptive Learning Rates
4. Regularization for Deep models
  - Model Selection
  - Underfitting, and Overfitting
  - L1 and L2 Regularization
  - Dropout
  - Challenges - Vanishing and Exploding Gradients, Covariance shift
  - Parameter Initialization
  - Batch Normalization
5. Convolutional Networks
  - Convolutions for Images
  - Learning a Kernel
  - Padding and stride, Channels, Pooling
  - Design of CNN
  - Popular CNN architectures
  - Transfer Learning
  - Applications of CNN
6. Sequence Models
  - Recurrent Neural Networks
  - Back-propagation through time
  - Challenge - Exploding - Vanishing gradient and Gates
  - Popular RNN architectures
  - Applications of RNNs
7. Attention Mechanism
  - Attention Pooling
  - Attention Scoring Functions
  - Multi-Head Attention, Self-Attention and Positional Encoding
  - Transformer architecture
  - Applications of Transformers
8. Neural Network search
  - Search Space
  - Search algorithms
  - Evaluation Strategy
9. Time series Modelling and Forecasting
  - Univariate, Multivariate and Multi-step CNN Models
  - Univariate, Multivariate and Multi-step LSTM Models





#### 10. Other Learning Techniques

- Federated learning
- Meta learning
- Online (incremental) learning

#### Part B: Learning Plan

| Contact Session | List of Topic Title                     | Sub-Topics                                                                                                                                                                                                                                                                                                                                                                          | Reference                                                                                                                                                                                                                                                                                                                              |
|-----------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1               | <b>Fundamentals of Neural Network</b>   | <ul style="list-style-type: none"> <li>• Supervised, unsupervised, semi-supervised and reinforcement learning problems.</li> <li>• Why Deep Learning?</li> <li>• Applications of Deep Learning</li> <li>• Biological neuron vs artificial neuron</li> <li>• Connectionism model</li> <li>• Perceptron</li> <li>• Perceptron learning algorithm</li> <li>• XOR Problem</li> </ul>    | T1 – Ch1<br><br><a href="http://mlsp.cs.cmu.edu/people/rsingh/docs/Chapter1_Introduction.pdf">http://mlsp.cs.cmu.edu/people/rsingh/docs/Chapter1_Introduction.pdf</a>                                                                                                                                                                  |
| 2               | <b>Fundamentals of Neural Network</b>   | <ul style="list-style-type: none"> <li>• Perceptron and Perceptron learning algorithm</li> <li>• Multilayer Perceptron (MLP)</li> <li>• MLP as classifiers and Universal approximators</li> <li>• Issue of Depth and Width</li> </ul>                                                                                                                                               | <a href="http://mlsp.cs.cmu.edu/people/rsingh/docs/Chapter1_Introduction.pdf">http://mlsp.cs.cmu.edu/people/rsingh/docs/Chapter1_Introduction.pdf</a><br><br><a href="http://mlsp.cs.cmu.edu/people/rsingh/docs/Chapter2_UniversalApproximators.pdf">http://mlsp.cs.cmu.edu/people/rsingh/docs/Chapter2_UniversalApproximators.pdf</a> |
| 3               | <b>Deep Feedforward Neural Networks</b> | <ul style="list-style-type: none"> <li>• Forward and backward propagation</li> <li>• Backward Propagation (review)</li> <li>• Training a DNN using Gradient Descent algorithm (review)</li> <li>• Computational Graphs (review as already discussed in MFML)</li> <li>• Activation Functions</li> <li>• Softmax Regression</li> <li>• Impact of depth in DNN</li> </ul>             | T1 – Ch4 and Ch3.4                                                                                                                                                                                                                                                                                                                     |
| 4               | <b>Optimization of Deep models</b>      | <ul style="list-style-type: none"> <li>• Challenges in Neural Network Optimization – saddle points and plateau</li> <li>• Non-convex optimization intuition</li> <li>• Stochastic Gradient Descent (SGD), Minibatch SGD</li> <li>• Overview of Rprop, Quickprop</li> <li>• Momentum, Nesterov's Accelerated Momentum</li> <li>• Algorithms with Adaptive Learning Rates,</li> </ul> | T1 – Ch11                                                                                                                                                                                                                                                                                                                              |





|    |                                              |                                                                                                                                                                                                                                                                                                      |                  |
|----|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
|    |                                              | Adagrad, RMSprop, ADAM                                                                                                                                                                                                                                                                               |                  |
| 5  | <b>Regularization for Deep models</b>        | <ul style="list-style-type: none"> <li>Model Selection</li> <li>Underfitting, and Overfitting</li> <li>L1 and L2 Regularization</li> <li>Dropout</li> <li>Challenges - Vanishing and Exploding Gradients, Covariance shift</li> <li>Parameter Initialization</li> <li>Batch Normalization</li> </ul> | T1 – Ch4 , 7.5   |
| 6  | <b>Convolutional Networks</b>                | <ul style="list-style-type: none"> <li>Convolutions for Images</li> <li>Learning a Kernel</li> <li>Padding and stride, Channels, Pooling</li> <li>Design of CNN</li> </ul>                                                                                                                           | T1 – Ch6, Ch7    |
| 7  | <b>Convolutional Networks</b>                | <ul style="list-style-type: none"> <li>Popular CNN architectures</li> <li>Transfer Learning</li> <li>Applications of CNN</li> </ul>                                                                                                                                                                  | T1 – Ch6, Ch7    |
| 8  | <b>Review</b>                                | Review of contact session 1 to 7                                                                                                                                                                                                                                                                     |                  |
| 9  | <b>Sequence Models</b>                       | <ul style="list-style-type: none"> <li>Recurrent Neural Networks</li> <li>Types of Sequences and RNNs</li> <li>Back-propagation through time</li> <li>Challenge - Exploding - Vanishing gradient and Gates</li> </ul>                                                                                | T1 – Ch8, Ch9    |
| 10 | <b>Sequence Models</b>                       | <ul style="list-style-type: none"> <li>Popular RNN architectures - GRU, LSTM, BiLSTM</li> <li>Applications of RNNs</li> </ul>                                                                                                                                                                        | T1 – Ch8, Ch9    |
| 11 | <b>Attention Mechanism</b>                   | <ul style="list-style-type: none"> <li>Attention Pooling</li> <li>Attention Scoring Functions</li> <li>Multi-Head Attention, Self-Attention and Positional Encoding</li> </ul>                                                                                                                       | T1 – Ch10        |
| 12 | <b>Attention Mechanism</b>                   | <ul style="list-style-type: none"> <li>Transformer architecture</li> <li>Applications of Transformers</li> </ul>                                                                                                                                                                                     | T1 – Ch10        |
| 13 | <b>Neural Network search</b>                 | <ul style="list-style-type: none"> <li>Search Space</li> <li>Search algorithms</li> <li>Evaluation Strategy</li> </ul>                                                                                                                                                                               | R5               |
| 14 | <b>Time series Modelling and Forecasting</b> | <ul style="list-style-type: none"> <li>Univariate, Multivariate and Multi-step CNN Models</li> <li>Univariate, Multivariate and Multi-step</li> </ul>                                                                                                                                                | R4<br>Ch-8, Ch-9 |





|    |                                  |                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |                                  | LSTM Models                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 15 | <b>Other Learning Techniques</b> | <ul style="list-style-type: none"> <li>Federated learning</li> <li>Meta learning</li> <li>Online (incremental) learning</li> </ul> | <a href="https://arxiv.org/pdf/1902.04885.pdf">https://arxiv.org/pdf/1902.04885.pdf</a><br><a href="https://arxiv.org/pdf/1903.10635.pdf">https://arxiv.org/pdf/1903.10635.pdf</a><br><a href="https://arxiv.org/pdf/2004.05439.pdf">https://arxiv.org/pdf/2004.05439.pdf</a><br><a href="https://arxiv.org/pdf/1802.07569.pdf">https://arxiv.org/pdf/1802.07569.pdf</a><br><a href="https://arxiv.org/pdf/1904.07734v1.pdf">https://arxiv.org/pdf/1904.07734v1.pdf</a> |
| 16 | <b>Review</b>                    | Review of contact session 9 to 15                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

### Experiential Learning Components:

1. Lab work: 7
2. Project work: 0
3. Case Study: 0
4. Simulation: 0
5. Work Integrated Learning Assignment- 2 Assignments
6. Design work/ Field work: 0

### Objective of Experiential Learning Component:

Hands on sessions on implementation of

- a) MLP
- b) CNN
- c) RNN
- d) Transformer

### Scope of Experiential Learning Component:

**Programming language:** Python

**Tools and libraries:** Jupyter, Numpy, Scipy, Pandas, ScikitLearn, Matplotlib, Seaborn, Tensorflow, Pytorch, Keras

### Lab Infrastructure:

Online / Open source / Virtual lab / Google Colab

### List of Experiments:

| Lab No | Lab Objective                                 | Session Reference |
|--------|-----------------------------------------------|-------------------|
| 1      | Introduction to Tensorflow, Keras             | 2                 |
| 2      | Computational graph in Pytorch                | 3                 |
| 3      | Deep Neural Network with Back-propagation and | 4                 |





|   |                         |    |
|---|-------------------------|----|
|   | optimization            |    |
| 4 | CNN                     | 6  |
| 5 | RNN, LSTM               | 9  |
| 6 | Transformers            | 12 |
| 7 | Time series forecasting | 15 |

### **Evaluation Scheme:**

**Legend:** EC = Evaluation Component; AN = After Noon Session; FN = Fore Noon Session

| Evaluation Component | Name (Quiz, Lab, Project, Mid-term exam, End semester exam, etc.) | Type (Open book, Closed book, Online, etc.) | Weight | Duration  | Day, Date, Session, Time |
|----------------------|-------------------------------------------------------------------|---------------------------------------------|--------|-----------|--------------------------|
| EC – 1*              | Quiz                                                              | Online                                      | 10%    | 1 week    | To be announced          |
|                      | Assignment/Lab Assignment / Lab Exams                             | Online                                      | 20 %   | 10 days   | To be announced          |
| EC - 2               | Mid-Semester Test                                                 | Closed Book                                 | 30%    | 2 hours   | To be announced          |
| EC - 3               | Comprehensive Exam                                                | Open Book                                   | 40%    | 2 ½ Hours | To be announced          |

EC1\* (20% - 30%): Quiz (optional): 5-10 %, Lab Assignment/Assignment: 20% - 30%

Syllabus for Mid-Semester Test (Closed Book): Topics in Contact session: 1 to 8

Syllabus for Comprehensive Exam (Open Book): All topics

### **Important Links and Information:**

**eLearn Portal:** <https://elearn.bits-pilani.ac.in>

Students must visit the eLearn portal regularly and stay updated with the latest announcements and deadlines.

**Contact Sessions:** Students should attend the online lectures as per the schedule provided on the eLearn portal.

### **Evaluation Guidelines:**

- EC-1 consists of either two Assignments or three Quizzes. Students will attempt them through the course pages on the eLearn portal. Announcements will be made on the portal in a timely manner.
- For Closed Book tests: No books or reference material of any kind will be permitted.
- For Open Book exams: “open book” means text/ reference books (publisher copy only) and does not include any other learning material. No other learning material will be permitted during the open book examinations. For Detailed Guidelines refer to the attached document.  
[EC3 Guidelines](#)
- If a student is unable to appear for the Regular Test/Exam due to genuine exigencies, the student should follow the procedure to apply for the Make-Up Test/Exam, which will be made available on the eLearn portal. The Make-Up Test/Exam will be conducted only at selected exam centres on the dates to be announced later.

It shall be the responsibility of the individual student to be regular in maintaining the self-study schedule as given in the course handout, attend the online lectures, and take all the prescribed evaluation components such as Assignments/Quizzes, Mid-Semester Tests and Comprehensive Exams according to the evaluation scheme provided in the handout.

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